# First Pair Press Founding Documents

The manifesto and the Bell Labs publishing workflow

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First Pair Press

# *I. The First Pair Bell Labs Manifesto*

## *We Believe Books Are Software*

Books are among humanity's oldest technologies.

Software is among its newest.

Both are systems for encoding ideas.

Software has spent decades learning how to evolve. Books have largely learned how to imitate magazines. We believe books deserve better.

A book should not be a static artifact. It should be a living system whose history, sources, reasoning, and construction are as inspectable as a great piece of software.

## *Bell Labs Was Never About Telephones*

People remember Bell Labs for the transistor, Unix, information theory, C, and laser physics.

The deeper lesson is that Bell Labs built tools that amplified human thought.

Unix was not merely an operating system. It was an operating philosophy.

- Write programs that do one thing well.
- Connect them together.
- Keep interfaces simple.
- Prefer text over opaque formats.
- Trust composition over monoliths.

Publishing deserves the same philosophy.

## *The Book Is Semantic Source*

The manuscript is not the final product.

It is semantic source.

Publishing is compilation.

Typography is only one backend among many.

The author's job is to express meaning. The toolchain's job is to render that meaning beautifully.

Source should be:

- plain text;
- readable without proprietary software;
- version controlled;
- semantically rich;

- durable for decades.

Presentation should always be derived.

### *AI Is Another Unix Filter*

Artificial intelligence should not replace thinking.

It should replace friction.

Like every good Unix tool, AI performs one task in a larger pipeline.

It can:
- discover references;
- verify quotations;
- propose diagrams;
- generate indexes;
- check consistency;
- compare editions;
- summarize research;
- expose uncertainty.

Judgment remains human.

AI is an assistant, never an authority.

### *Every Book Should Be Reproducible*

Books should build like software.

Given the same manuscript, references, assets, and toolchain, the same book should emerge.

No hidden formatting.

No invisible edits.

No proprietary magic.

Publishing should become reproducible research.

### *Plain Text Is Civilization's Longest-Lived File Format*

Stone.

Clay.

Paper.

ASCII.

Unicode.

Plain text survives revolutions in technology.

We choose formats that our grandchildren can still read.

### *Many Renderers, One Source*

A manuscript should outlive its rendering engine.

Today's beautiful formatter may someday disappear.

Meaning must not.

From one semantic source we should be able to produce:

- Typst;
- troff;
- HTML;
- EPUB;
- PDF;
- print;
- presentations;
- future formats not yet invented.

Renderers compete.

Source endures.

### *Pandoc Is Our Linker*

Compilers transformed software engineering.

Pandoc can play a similar role in publishing.

It links semantic source to multiple rendering engines without coupling authors to any single implementation.

The publishing pipeline becomes:

```
Semantic Source
|
v
AI Enrichment
|
v
Reference Resolution
|
v
Diagram Generation
|
v
Pandoc
|
v
```

Typst / troff / HTML / EPUB / Print

The intermediate representation matters more than any single renderer.

### *Troff and Typst*

Troff represents the Bell Labs tradition.

Typst represents a modern reimagining of programmable typography.

Neither is the destination.

Both are excellent backends.

First Pair Press is committed to ideas, not tools.

We choose the renderer that best serves the reader.

### *Diagrams Are Programs*

Whenever possible, diagrams should be generated from source.

Relationships become graphs.

Timelines become events.

Maps become geometry.

Figures become executable knowledge.

### *References Are First-Class Citizens*

Every important claim deserves provenance.

Citations should explain not only where information came from, but why it matters.

Knowledge should be inspectable.

### *We Publish Reasoning*

The internet optimizes for conclusions.

We optimize for reasoning.

Readers should understand not only what we believe, but how we arrived there.

Good books teach methods of thought.

### AI-Native Does Not Mean AI-Written

AI-native publishing means embracing computational thinking throughout research, editing, verification, accessibility, translation, indexing, and production.

Human authors remain responsible for ideas.

Machines amplify craftsmanship.

### Small Tools, Loose Coupling

One tool researches.

One verifies.

One builds indexes.

One renders Typst.

One renders troff.

One generates EPUB.

Each tool should be replaceable.

The pipeline is more important than any component.

### Books Should Age Gracefully

Every edition should preserve its history.

Every correction should be visible.

Every revision should be traceable.

Knowledge evolves.

Books should evolve with it.

### The Library Is a Knowledge Graph

Books should not live in isolation.

Ideas, people, places, events, and concepts should connect across the catalog.

A publishing house should become an evolving graph of knowledge.

### First Pair

The name "First Pair" has two meanings.

The first pair of programmers.

The first pair of collaborators.

Human and machine.

Neither replaces the other.

Together they build something neither could create alone.

### *The Bell Labs of Books*

Our ambition is larger than publishing books.

We seek to build the Bell Labs of books.

A place where writers, researchers, designers, programmers, historians, and intelligent machines create better tools for thinking.

Books are not the final product.

They are the visible proof that the system works.

### *Our Principles*

We choose plain text over proprietary formats.

We choose semantic structure over visual markup.

We choose reproducible builds over manual layout.

We choose composable pipelines over monolithic software.

We choose open standards over lock-in.

We choose transparency over mystique.

We choose reasoning over rhetoric.

We choose collaboration over automation.

We choose enduring ideas over fashionable tools.

We are committed not to troff.

Not to Typst.

Not to Pandoc.

We are committed to a philosophy:

> Ideas should outlive their implementations.

Tools should remain replaceable.

Knowledge should remain permanent.

That is the Bell Labs way.

That is the First Pair way.

### *The Proof Must Build*

A manifesto is not enough.

First Pair Press must prove its ideas in working books.

The first proof lives in this repository as ‘proofs/kiffness-loop-lab‘. It takes a small music-manual idea and builds it as a reproducible human-AI publishing artifact:

- plain Markdown source;
- AI collaboration notes;
- Pandoc conversion;
- Typst PDF;
- EPUB;
- Neatroff PDF;
- groff fallback PDF;
- utmac troff source and PDF;
- a VERSION.md artifact manifest.

The proof is deliberately modest. It is small enough to inspect and real enough to break.

That is where trust begins.

# II. First Pair Bell Labs

## *The Bell Labs Publishing Philosophy*

The original Bell Labs publishing pipeline was not "troff" so much as a philosophy:

> Write plain text. Everything else is a program.

Every stage was a filter.

```
text
 |
 +-- refer   (bibliography)
 |
 +-- pic     (diagrams)
 |
 +-- tbl     (tables)
 |
 +-- eqn     (math)
 |
 +-- troff
 |
 +-- postscript
 |
 PDF
```

The beauty is that every stage is optional.

## *A Bell Labs Book*

```
.TL
Lighthouse Republics

.AU
Alexy Khrabrov

.AI
First Pair Press

.AB
A short history of the United States and Venice together,
and their eternal afterlife.
.AE

.SH
The Republic
```

```
Venice was never merely a city.

It was an operating system.

Its abstractions outlived its hardware.

.PP

America inherited many of them.

.IP •
Federalism

.IP •
Commerce

.IP •
Constitutional abstraction

.PP

These ideas are older than nations.
```

No XML. No YAML. No Markdown. No HTML. Just text.

### *Figures*

Instead of drawing SVGs, write programs.

```
box "Republic"

arrow

box "Empire"

arrow

box "Nation"

arrow

box "Cloud"
```

Produces:

```
+---------+
|Republic |
+---------+
     |
     V
```

```
+--------+
|Empire  |
+--------+
     |
     V
+--------+
|Nation  |
+--------+
     |
     V
+-------+
|Cloud  |
+-------+
```

pic is programmable:

```
define lighthouse {
   circle radius .08
   line up .3
   box invis ht .2 wid .2
}

lighthouse

move right 1

lighthouse

arrow
```

### *Tables*

```
.TS
center box;

l l l.

Republic   Founded   Fate
Venice     697       Endured
USA        1776      Ongoing
.TE
```

### *Equations*

```
.EQ

Liberty ~= Order + Commerce

.EN
```

or:

```
.EQ

sum from i=1 to n x sub i

.EN
```

### *References*

```
.[
VeniceHistory
.]
```

Then:

```
refer references.bib
```

Automatic numbering.

### *Version Control*

Everything is plain text.

```
-America inherited Rome.
+America inherited Venice.
```

instead of:

```
Binary file changed.
```

### *Build System*

```
book.pdf: book.ms refs
    refer refs |
    pic |
    tbl |
    eqn |
    troff -ms |
    ps2pdf > book.pdf
```

### *Why UNIX People Loved This*

Each tool does one thing well:

- eqn -- equations;
- tbl -- tables;

- pic -- drawings;
- troff -- layout.

No component knows anything about another.

### *Bell Labs Heritage*

The original The C Programming Language and many editions of The UNIX Programming Environment were produced with ms macros.

Brian Kernighan famously preferred writing text over manipulating formatting.

The markup disappears while you are writing.

### *A Vision for First Pair Press*

Rather than merely reviving troff, modernize the Bell Labs pipeline around AI.

```
Markdown / Troff hybrid
    |
    v
AI semantic pass
    |
    v
refer
    |
    v
pic (AI-generated diagrams)
    |
    v
tbl
    |
    v
eqn
    |
    v
troff
    |
    v
PDF
    |
    +-- EPUB
    +-- HTML
    +-- Print
```

The AI enriches the document with references, diagrams, indexes, glossary entries, and consistency checks before handing a semantically rich document to a deterministic typesetting engine.

### *First Pair Macros*

```
.FP.TITLE "Lighthouse Republics"

.FP.AUTHOR "Alexy Khrabrov"

.FP.EPIGRAPH
The sea remembers every republic.
.FP.END

.FP.CHAPTER "The Lagoon"

The lagoon was Venice’s first compiler.

.FP.QUOTE
Cities survive by changing their abstractions.
.FP.END

.FP.SIDEBAR "Why Lagoons Matter"
The lagoon acted as both moat and marketplace...
.FP.END

.FP.AI "Research Notes"
Summarize recent scholarship on Venetian naval logistics.
.FP.END
```

The ‘.FP.AI‘ blocks exist only in source. During the build they are resolved into finished prose or omitted entirely, leaving the published book free of AI scaffolding while preserving the Bell Labs ideal of simple, readable source.

### *Core Idea*

The combination of classic troff composability with AI-aware semantic macros could become the defining publishing system for First Pair Press: AI-native books built on timeless UNIX principles.

### *The Concrete First Pair Workflow*

The proof workflow now lives in this repo rather than only in the manifesto:

```
publishing/PUBLISH.md
publishing/scripts/build-firstpair-book.sh
publishing/scripts/md-to-utmac.py
publishing/scripts/setup-utmac.sh
publishing/tmac/fp.tmac
proofs/kiffness-loop-lab/
proofs/kiffness-loop-lab/source.fp.tr
```

It uses the current local book practice from QueryGraph and usavenice:

- Pandoc links semantic Markdown to every renderer.
- Typst builds the modern reader PDF and EPUB.
- Pandoc ms builds the groff fallback source.

- A hand-authored ‘.FP.*‘ troff source builds the pure First Pair/Bell Labs proof through ‘fp.tmac‘, utmac, and Neatroff.
- Neatroff, from ‘~/src/neatroff_make‘, builds the Bell Labs PDF from the generated utmac source.
- groff remains a compatibility fallback because the usavenice shipping pipeline already proves it at book scale.
- utmac adds a richer macro vocabulary for metadata, headings, notes, summaries, and book-like troff source.

The proof book is intentionally small: ‘proofs/kiffness-loop-lab‘. It is derived from the Kiffness controller-manual workflow, but its main purpose is not music instruction. Its purpose is to prove that a human-readable source can produce several deterministic book artifacts while leaving enough evidence for the next human or AI collaborator to understand what happened.

## *What Builds Today*

Run:

```
proofs/kiffness-loop-lab/build.sh
```

The build emits:

```
firstpair-loop-lab-typst.pdf
firstpair-loop-lab-typst.epub
firstpair-loop-lab-firstpair.pdf
firstpair-loop-lab-firstpair.tr
firstpair-loop-lab-neatroff.pdf
firstpair-loop-lab-groff.pdf
firstpair-loop-lab-utmac.pdf
firstpair-loop-lab-utmac.tr
VERSION.md
```

‘firstpair-loop-lab-firstpair.pdf‘ is the source-level implementation of the macro sketch above. It is built from ‘source.fp.tr‘, a hand-authored troff file using ‘.FP.TITLE‘, ‘.FP.CHAPTER‘, ‘.FP.ITEM‘, ‘.FP.SIDEBAR‘, and ‘.FP.AI‘. The ‘.FP.AI‘ blocks remain visible in source and are omitted from the rendered PDF.

The utmac path is real, not merely a notation experiment. The setup script fetches the current utmac upstream into ‘.tools/utmac‘, runs its makefile so generated macros such as ‘u-idx.tmac‘ and ‘u-ref.tmac‘ exist, and then supports both the hand-authored ‘.FP.*‘ source and the Markdown-derived ‘.tr‘ source through Neatroff.

## *Human-AI Source Discipline*

The AI should not be hidden in the result.

It should be visible in the workshop:
- AI.md says how future agents should reason about the proof.
- manuscript.md remains readable by a person.
- source.fp.tr remains readable as direct Bell Labs source.

- publishing/tmac/fp.tmac is the inspectable First Pair macro layer.
- generated ms and tr files remain inspectable.
- VERSION.md records artifact names and page counts.
- renderer warnings are preserved as logs.

The published book should be clean. The build room should be honest.

That is the actual First Pair: the human sets intent, the AI removes friction, and the text remains durable enough for either one to return.